

XML Parsing Error: prefix not bound to a namespace

Location: file:///E:/LFY/LFY%202010/LFY%20Aug%202010/firewall_Manishrane/Figure%201.1%20Actual%20Scenario.svg
Line Number 186, Column 4:

```
<use xlink:href="#lend5" transform="scale(1.6176470588235)"/>
```

Figure 1: Actual scenario

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Location: file:///E:/LFY/LFY%202010/LFY%20Aug%202010/firewall_Manishrane/Figure%201.2%20Faults%20Tolerant%20Case.svg
Line Number 189, Column 4:

```
<use xlink:href="#lend5" transform="scale(1.6176470588235)"/>
```

Figure 2: Fault tolerant case

communication) travelling across it. The firewall is a device that differentiates genuine packets for different types of connections. It filters the traffic not only based on the port definitions, but based on the packet state as well.

In our scenario, we will be using the following components to build a stateful HA firewall cluster:

iptables: to perform packet filtering

keepalived: to work as a VRRP (Virtual Router Redundancy Protocol) solution; we've chosen it to automate the takeover process of the virtual IPs

conntrack: to maintain the connection/packet state.

fwbuilder: to build, compile and push the policy.

Fedora 12: to be used as a firewall platform.

In our scenario, we have the following topology (also see Figure 1):

Interface name	Firewall1	Firewall2	Cluster IP or VRRP IP
eth0	192.168.170.252/24	192.168.170.253/24	192.168.170.254/24
eth1/Sync Interface	172.16.1.252/24	172.16.1.253/24	—
eth2	10.1.1.252/24	10.1.1.253/24	10.1.1.254/24

Initially, we consider that Firewall2 holds the virtual IPs that are in bold font in Figure 1. The deployed settings follow the classical primary/backup approach, where Firewall2 is active, and Firewall1 waits for the failure of the active host.

Once Firewall2 experiences a failure, Firewall1 takes over the virtual IPs, and traffic will opt for the secondary path without dropping established connections.

Installation of firewall components

Iptables is used to set up firewall rules (or generally called access lists). Several different tables may be defined, which contain a number of built-in chains. Every chain is a list of rules. A rule consists of conditions which can match a set of

packets. *Iptables* features:

- Stateless packet filtering (IPv4 and IPv6)
- Stateful packet filtering (IPv4 and IPv6)
- All kinds of network addresses and port translation, e.g., NAT/NAPT (IPv4 only)
- Flexible and extensible infrastructure
- Multiple layers of APIs for third party extensions

To add or edit *iptables* firewall policies, you need to have root-equivalent privileges.

On nearly all Linux systems, *iptables* is installed as */usr/sbin/iptables* and you can refer to the documentation in its man pages using the command *man iptables*.

Keepalived provides a strong and robust health-checking mechanism for clusters. The main goal of the *keepalived* project is to add a strong and robust VRRP facility to the HA cluster. *Keepalived* is entirely written in C, and handles fail-overs seamlessly, using the VRRPv2 protocol. It can also be used to implement load-balancing (to divide the load across multiple servers for the best performance). However, in our scenario, we will focus on using it for high availability.

You can download *keepalived* from its homepage, <http://www.keepalived.org/>. In order to compile, *keepalived* needs the following libraries installed: OpenSSL (www.openssl.org) and *popt*. I am using *keepalived* version 1.1.20 for this project. Download the compressed file from the *Download* section, and untar it in a separate folder under */opt*, where we will be downloading these components.

```
[root@Firewall1 ~]# mkdir /opt/hacluster
[root@Firewall1 ~]# cd /opt/hacluster/
[root@Firewall1 hacluster]# tar -zxvf keepalived-1.1.20.tar.gz
[root@Firewall1 hacluster]# cd keepalived-1.1.20
[root@Firewall1 keepalived-1.1.20]# ./configure && make && make install
```

If everything goes correctly, *keepalived* will be installed under */usr/local/*. Go to the *etc* directory under */usr/local/*, and copy the files to an appropriate system path.